High School Quality and Subway Access in New York City

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While other cities send their teens to school on school buses or personal vehicles, New York City gives them OMNY cards. NYC public high school students receive free subway or bus rides on school days in order to commute to and from school (DOE transportation eligibility). Though it is well known that there is great socioeconomic and spatial disparity in subway access in NYC (eg. Baghestani et al., 2025), it is yet unknown if schools are also disparately placed near higher quality transportation options based on school metrics. If high-output schools are closer to higher quality public transportation, then that is a form of systemic inequity between students at elite public high schools and those at low performing schools. It is possible that the disparity between schools can create a feedback loop for students, such that those who dislike school perform poorly, go to lower rated schools that are not located near convenient public transportation, and then dislike going school more because it is inconvenient. However it is possible that high-achieving students choose schools that are convenient, or that well-funded schools are built near transit hubs. In this study, I investigate the spatial relationship between subway access and school quality. Is school quality, as measured by student performance, predicted by ease of access to the subway? This is an important policy question, because it seeks to measure inequity as a function of the built environment.

School location has long been linked to quality worldwide. In an analysis of British school output in comparison to nearby economic conditions, Conduit et al. (1996) found that local home ownership (along with other economic variables) was a powerful predictor of school achievement. At the same time, school output may also have a spillover effect on the rest of the community. Figlio and Lucas (2004) studied the effect of Florida schools' quality ratings on nearby housing prices, finding an independent effect of grade on the nearby housing market. In other words school rating may have a tangible effect on the built environment around it. Thus, it

is possible that school quality either has an effect on how transportation is built around it, or that top schools are built where they are most accessible. At the same time, NYC is unique in many ways, most relevantly for its extensive school choice policies. Schwartz et al. (2014) found that while zoned school performance is similarly linked to NYC housing values, school choice policies may "break the link;" the opening of "choice schools" in expensive neighborhoods with top zoned schools lowers housing demand in those neighborhoods. The rise of choice schools makes it all the more necessary to understand the link between transportation proximity and school achievement.

These same school choice policies that decouple housing prices from school quality also tend to silo students based on academic achievement (Nathanson et al., 2013). A report from the Institute for Educational and Social Policy on the school placement of low-achieving students found that students tend to prefer schools that are close to home or convenient, and poor students tend to also be lower-achieving than wealthier peers. This means that low-achieving students and schools tend to be concentrated in poor neighborhoods (Nathanson et al., 2013). If NYC has disparities in transportation access, and there are disparities in high school quality and location linked to social class and academic achievement, then it stands to follow that academic output for a given school may be linked to the quality of transportation available nearby for students.

One 2018 report on pupil transportation in NYC from the Urban Institute used student-level data to examine student commute patterns in relation to school quality (Cordes & Schwartz, 2018). In other words, do students who use public transportation go to better schools? They found that while students who attend "choice schools" (ie. schools not their zoned school) do tend to go to schools with better output, students who attend "choice schools" tend to have lower-quality zoned schools, making travel a necessity for a better education. Furthermore,

students who used a school bus tended to have higher proficiency rates than those who used a MetroCard (Cordes & Schwartz, 2018). The difference between school bus rides and public transportation riders is of particular interest. According to the DOE transportation guide, students attending a regular public school are assigned a yellow school bus if they live in the same district as their school, while those who attend a non-public, charter, or gifted and talented program get a bus if they are within the same borough. Can Cordes and Schwartz (2018)'s findings be attributed to the buses being full of charter school and G&T students, or does the convenience of having a bus enable students to go to better schools? Moreover, their study only analyzed students in elementary school. Does the relationship between transportation and school outcomes hold for older students? These questions lie at the core of the present study.

Hypotheses and variables

I have two hypotheses. First, that schools nearer to subway stations will have higher performance scores, and second, that schools that have a greater variety of nearby subway lines (eg**.** more lines within a .5 mi radius) will have higher performance scores.

My dependent variable is performance score, a composite measure of school output in comparison to city averages, and my independent variables include the distance from each school to its nearest subway station, the number of unique subway lines within .5 mile and one mile radii, and demographic variables such as the percentage of students with IEPs, percentage of English Language Learners (ELL), and child poverty rates and median income in the school's census tract.

Data Sources

For my spatial data, I used the 2024 School point location and MTA Subways station point shapefiles from NYC Open Data, as well as the 2023 NY Census tracts data from

TIGER/line. For demographic data, I used Social Explorer to get data from the 2023 5-year estimate of the American Community Survey (ACS), which is conducted by the US Census. ACS variables include median household income, number of children living under the poverty line per census tract, and total population per tract. My school outcomes data came from the NYC Department of Education 2023-2024 School Quality Report (SQR), which tracks outcomes at the individual school level. This dataset includes my dependent variable, school performance score, a composite variable derived from measuring school performance against citywide averages. Per the DOE, factors that go into this score include ELA and Algebra regents scores, graduation rates, and post-secondary enrollment rates.

Data cleaning and processing

In R, I filtered schools in the NYC spatial school data to only include those on the 2023/4 SQR report (eg. High schools) and filtered the New York State census tract data to only include NYC tracts prior to export. I used standardization measures such as ensuring GEO_ID and other identifier variables were stored as a character across files, trimming variable names, changing case to uppercase, removing whitespace, and storing cleaned identifiers as new variables to avoid overwriting the original. All layers were reprojected to CRS 3857, and I converted the transit csv to an SF file. Missing subway line data was removed from the dataset, and variable names were later trimmed in preparation for file export; only the variables planned for use were exported.

Spatial joins and techniques:

In R, Census tract and demographic characteristics were spatially joined to school locations and school output data. I created buffers of .5 and 1mile to count unique train lines near each school (join type = Intersects); schools with no subway lines within each buffer zone counted as 0. Routes were originally entered as names (eg. Q R S); routes were separated and

each distinct line counted. I also created a variable measuring the distance from each school to the nearest subway station. I primarily used left-joins to maintain the integrity of the data (eg. preserve full list of schools when the joined dataset was missing data). In QGIS, I later performed a spatial join (using 'join attributes by location- summary') to add school information to tract polygons. This created count, mean, and median for each joined variable that were later used for mapping variables such as mean performance score per tract.

Data Limitations

I had originally planned to use data from the SAT, since that is an independent standardized test. Unfortunately, the most recent SAT data I could find was from 2012, making it too outdated to use here as my dependent variable. On a related note, the 2025 SQR was released in the middle of this project, making the 2023/2024 SQR a little outdated by contrast. I also do not have information related to average student commute time or method for each school, which could serve to separate schools that have a high percentage of commuters and thus have students that are more reliant on the quality public transportation compared to schools where students primarily live nearby.

GIS Methods and Software Used

Initial data cleaning and spatial joins were performed in R using packages sf, dplyr, readr, readxl, tmap, ggplot2, units, stringr, fuzzyjoin, and nngeo. Steps are described in greater detail above. A joined file consisting of school geometry point coordinates, GEO ID, SQR and ACS variables, variables calculating train line variety and distance from each school to its nearest train station as well as other tract geographic data was exported from R as a shapefile. A separate file containing NYC tract polygon and ACS data was exported as a geopackage. Distributions were mapped in QGIS; a spatial join of the two exported files was conducted using the 'join attributes

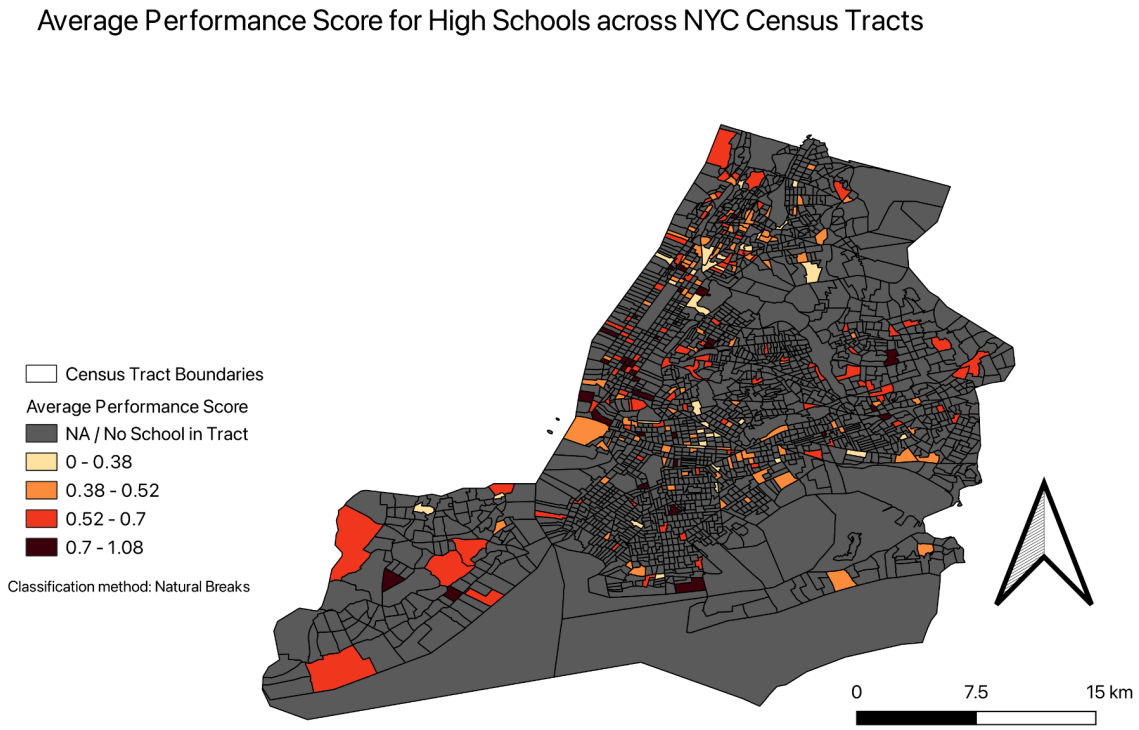
by location- summary' function to create a new layer, which was used for mapping tract level data in my chloroplath maps (See Appendix), while the points file was used to map school point-level data. This also created tract level averages that were used for mapping. As there are many tracts in NYC that did not have any high schools, an NA box on the legend was created by using the coalesce function in the layer properties symbology expression builder and assigning a negative number to the lowest class (eg. coalesce("PerfScr_mean",-1).

All spatial analysis was done in GeoDa. A school weights file was created for weighting, and used to calculate global and local Moran's I. Diagnostic tests of the model and an Ordinal Logistic Regression were performed in GeoDa, as were spatial error and spatial lag models that were later discarded based on the model's performance on diagnostics including the Lagrange Multiplier lag, error, and SARMA tests.

Results

Most NYC Census tracts did not contain a single high school; distribution of the average performance score for each NYC tract is mapped below (Map 1). Visually, there does not appear to be a strong pattern wherein one area of the city has all of the top schools. In line with the LISA analysis below, yellow (ie. very low performing schools) and orange (below average-scoring schools) appear to cluster together, as do red (above average) and dark red (very high performing) schools.

Map 1: Average High School Performance Scores of NYC Census Tracts

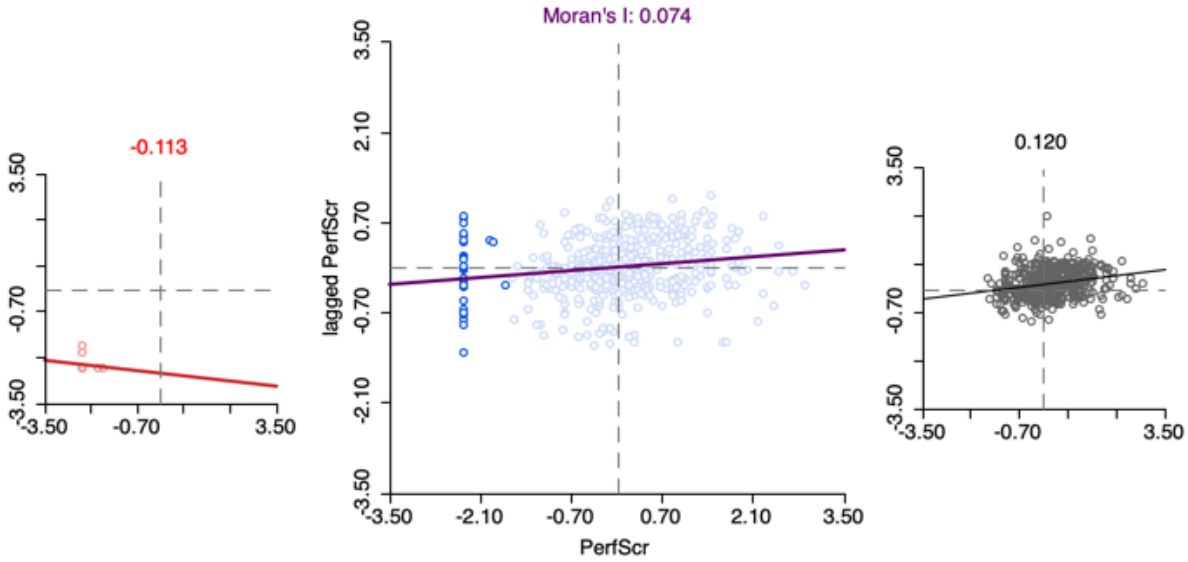


Global Moran's I

I first ran a global Moran's I test on performance score, to test if there was spatial autocorrelation. Each test was run with 999 permutations. The results indicate a modest positive spatial autocorrelation (I = .075, p=.001), with a left-hand tail indicating a significant group of low performance score outliers that exhibit less spatial clustering than the rest of the schools. When this tail was excluded from the Global Moran's I using regime regression, autocorrelation for performance score significantly increased (I=.12, p=.001). When isolated, those outlier schools displayed a negative spatial autocorrelation (I= -.113, p=.001), indicating that there may be a small number of low performing schools that follow a dispersion pattern of spatial

correlation (eg. located near high performing schools instead of clustering near other low-output schools).

Figure 1: Global Moran's. From left to right, results from the tail, overall, and without the tail



LISA

Next, I ran a Local Moran's I, also known as Local Indicators of Spatial Association (LISA) with 999 permutations using Monte Carlo simulation. The majority of results were not significant (n=395). At the 95% significance level, there were 41 High-High clusters, 31 Low-Low clusters, 23 Low-High, and 13 High-Low clusters (See Figure 2). When narrowed down to the 99% significance level, 468 results were not significant, and there were Nine High-High, 14 Low-Low, Seven Low-High, and Five High-Low clusters. This may mean that low-performing schools are more likely to cluster near each other, in line with the patterns of school attendance of low achieving students observed in Nathanson et al. (2013). Like with the Global Moran's I test, patterns of clustering and spatial autocorrelation around high school performance scores are modest, indicating less spatial segregation between high and low performing schools than expected.

Diagnostic Tests

The Jarque-Bera test of normality had a value of 221.39 (df= 2, p< 0.001), indicating that the data is not normally distributed. Both the Breusch-Pagan (X=91.2329, p<.001) and Koenker-Bassett estimators (X=41.6296, p < .001) of heteroskedasticity reject the null hypothesis, indicating that there is variance of the residuals present in the model. Of the six tests run for spatial dependence of the model, including Moran's I on model residuals and Lagrange Multiplier lag, error, and SARMA tests, none were significant, or even approaching significance (defined as .05 < X < .1; See Table 1 for complete values). Overall, there does not seem to be any statistically significant spatial dependence of the dependent variable or model residuals, indicating that an OLS model is appropriate for this data.

Table 1. Spatial Dependence Diagnostic Tests

Test	MI/DF	Statistic	p-value
Moran's I (error)	0.0236	1.496	0.135
Lagrange Multiplier (lag)	1	1.331	0.249
Robust LM (lag)	1	0.183	0.669
Lagrange Multiplier (error)	1	1.156	0.282
Robust LM (error)	1	0.007	0.932
Lagrange Multiplier (SARMA)	2	1.339	0.512

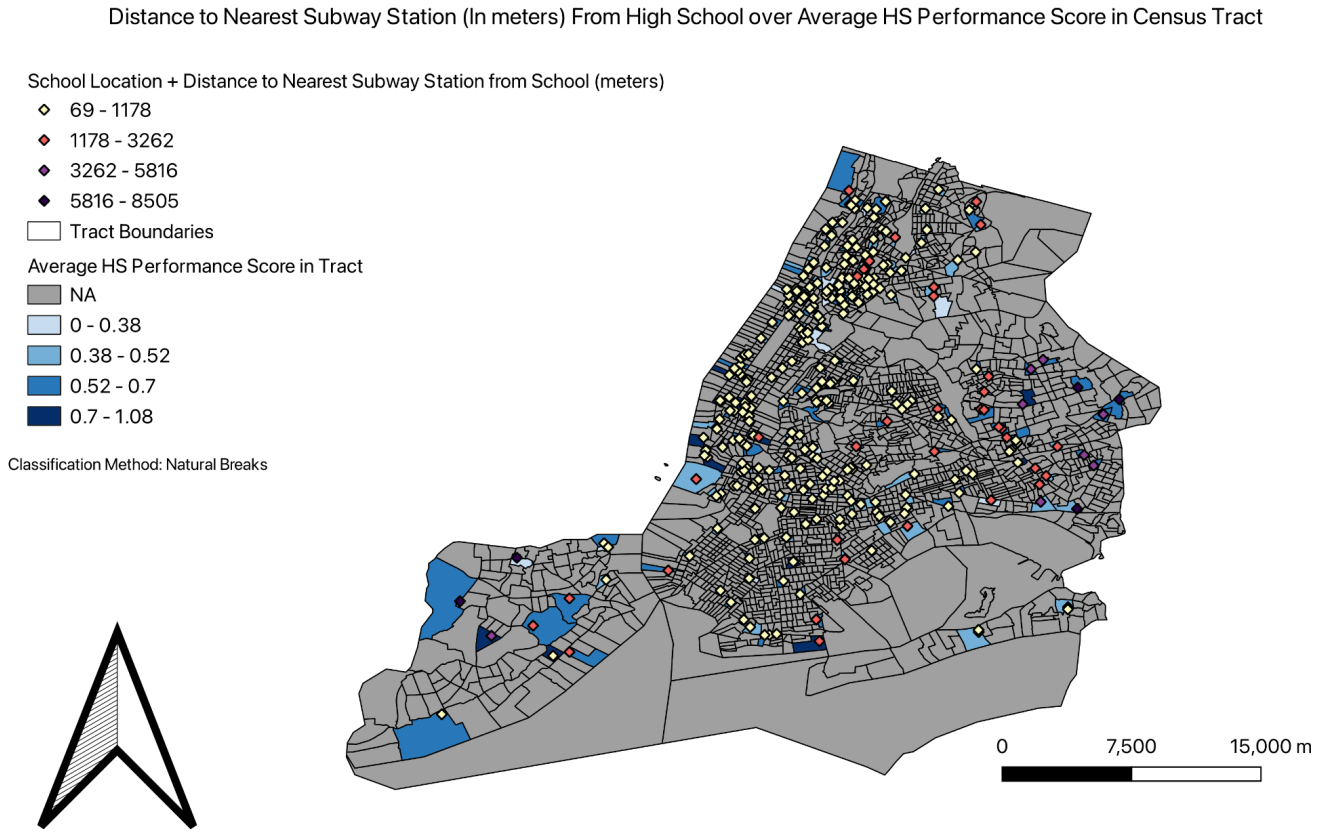
OLS

The Ordinary Least Squares (OLS) model had an adjusted R-squared of 0.28, (DF = 464, F= 23.498, P < .001).

Hypothesis 1

Hypotheses 1, that schools nearer to subway stations would have higher performance scores was tested using the nearest subway station variable calculated in R. Results were not statistically significant, with an extremely low coefficient (β = 1.1437e-05, SE = 7.17234e-06, t= 1.5946, p= 0.111), indicating that distance to nearest subway station was not a significant predictor of school performance score, disproving hypothesis 1. Map 2 shows that the vast majority of schools appear to be within 69 to 1178 meters from their nearest subway station, particularly in Manhattan and western Brooklyn, while some areas of the Bronx, eastern Queens, and Staten Island have clusters of schools that are much farther from the subway, though this does not appear to be correlated with performance.

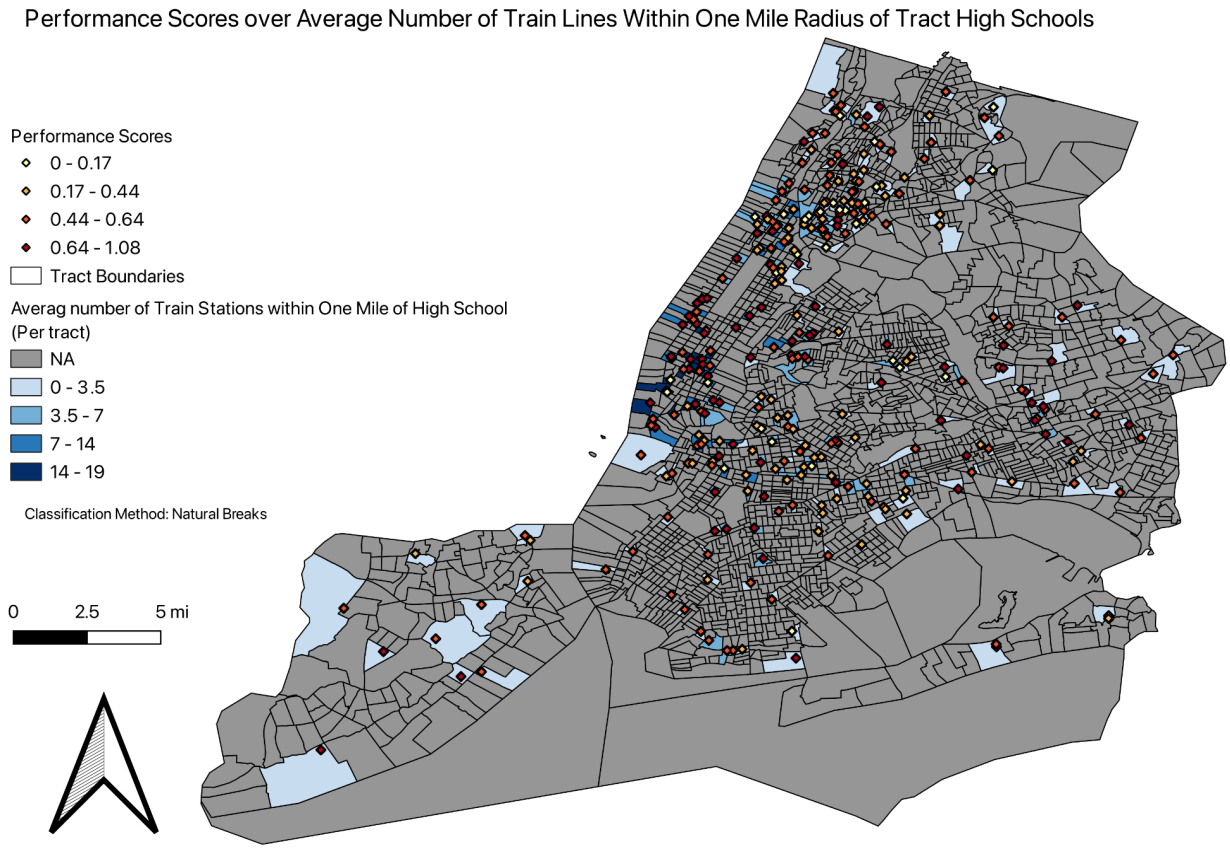
Map 2: Distance to Nearest Subway Station over Average High School Performance



Hypothesis 2

Hypothesis 2, that schools that have a greater variety of nearby subway lines will have higher performance scores, was operationalized by counting the number of unique subway lines within a .5 and one mile radius of the school. The half mile radius was not a significant predictor of performance score, though it could be argued it was approaching significance, and the coefficient was small and negative ($\beta = -0.008$, $SE = .004$, $t = -1.821$, $p = .069$). The number of stations within a one mile radius, however, was a significant positive predictor of school performance ($\beta = .008$, $SE = .003$, $t = 2.704$, $p = .007$), though the effect size was quite small. These findings suggest a scale-dependent effect of subway line variety, providing evidence in support of hypothesis 2. Map 3 shows performance scores mapped over the average number of train lines within a one mile radius of schools in each tract. Schools in lower Manhattan are by and large high performing and near many trains, while there doesn't appear to be much of a pattern elsewhere. This may account for the low coefficient size.

Map 3: High School Performance Scores over Average Number of Train Lines per School



Other variables

As one might expect, within-school demographic variables were significant predictors of school performance. Percentage of English Language Learners ($\beta = -0.482$, SE = 0.049, t = -9.869, p <.001) and percentage of students with IEPs ($\beta = -1.001$, SE = 0.109, t = -9.175, p <.001) were both strong negative predictors with large coefficients, while percent of positive instructor and learning environment ratings was a significant positive predictor ($\beta = 0.243615$, SE = 0.035, t = 6.923, p < 0.001). Surprisingly however, the median income (β = -9.87967e-09, SE = 2.02619e-07, t = -0.049, p = 0.961) and child poverty rate (β = -1.52691e-05, SE = 2.58926e-05, t = -0.5897, p = .556) of the schools' surrounding area were

NOT significant predictors of school performance. While both had a negative directionality, the coefficients of both variables were miniscule. See Table 2 for a full list of outcomes. The relationship between tract median income and school performance score is mapped below in Map 4. While the high income lower Manhattan area appears to also be densely populated with high performing high schools (with the exception of one very low performing school), the rest of the city appears to have no such spatial distribution. Low and medium income areas in western Queens appear to have several high performance schools, while the low-income south Brooklyn has several above-average high schools. At the same time, this map shows how low performing (light green) and high performing (dark green) tend to cluster near each other, not necessarily based on tract income.

Map 4: NYC High School Performance Scores over Median Income of Census Tract

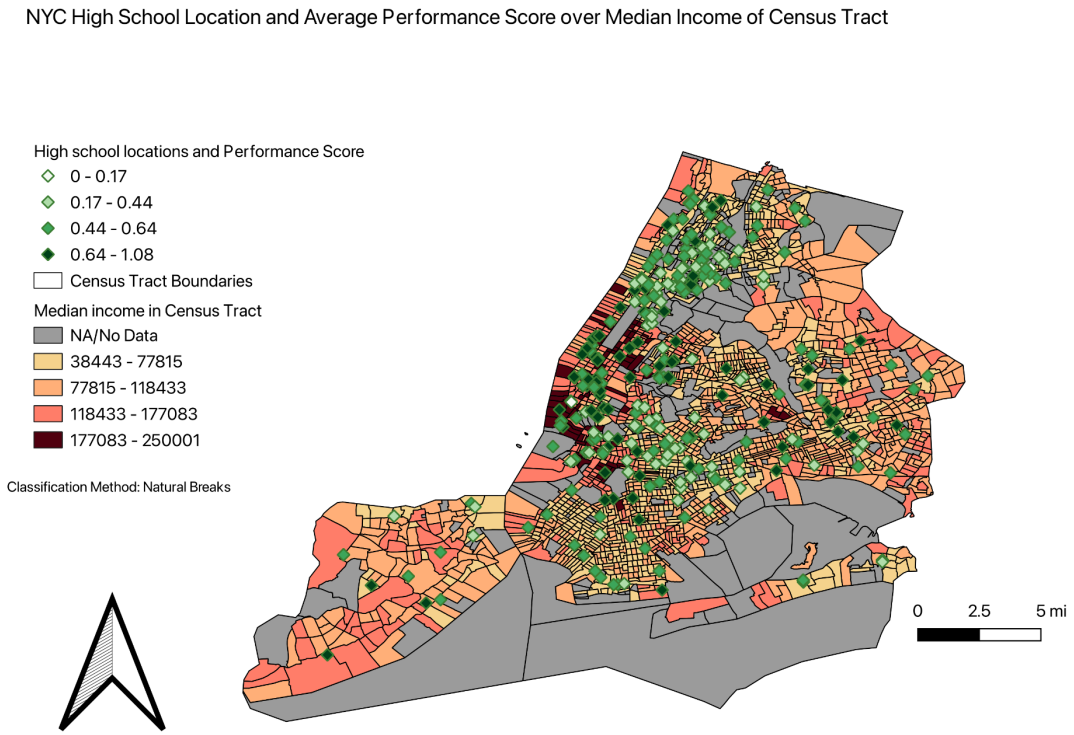


Table 2. Ordinary Least Squares (OLS) Regression Results

Variable	Coefficient	Std. Error	t-Statistic	p-value
Constant	0.563***	0.041	13.83	<0.001
Lines within 0.5 Mi	−0.008†	0.004	−1.82	0.069
Lines within 1 Mi	0.008**	0.003	2.70	0.007
ELL (%)	−0.482***	0.049	−9.87	<0.001
Child Poverty in Tract	−0.00002	0.00003	−0.59	0.556
IEP (%)	−1.001***	0.109	−9.18	<0.001
Median Income in Tract	−0.00000001	0.00000020	−0.05	0.961
Instruction/Environment % Positive	0.244***	0.035	6.92	<0.001
Distance to Nearest Station (m)	0.00001	0.00001	1.59	0.111

Discussion

Overall, findings indicate a scale dependent effect of subway variety (negative, insignificant when scale is narrow, positive and significant with broader radius) on NYC high school performance scores: However, it is nowhere near the effect of within-school demographic variables, though is to be expected that student characteristics would be dominant predictors of school performance. At the same time, census tract characteristics (e.g. median income and child poverty rates in the school's surrounding area) were not significant predictors of performance, with tiny effect sizes. This, combined with the lack of significance of subway distance, the heteroscedasticity, abnormality of distribution, and lack of spatial dependence of the overall

model indicate that school performance is more heavily tied to student characteristics than physical location.

Unlike Conduit et al. (1996), the socioeconomic factors in NYC schools' surrounding area was not a significant predictor of school quality. Combined with the finding that NYC high school performance has but a small link to subway accessibility, these findings support the idea that school choice can be a mechanism for breaking the link between education and the housing market, in line with Schwartz et al. (2014).

Visual inspection of the chloroplath maps also reveals that wealthy areas of lower Manhattan tend to have high performing schools and more train lines. It is possible that this area is largely responsible for the finding of subway line density as a weak, positive predictor of school performance. Moreover, there was a clear pattern in Map 2 such that schools located in the eastern peripheral areas of Queens tended to have significantly farther distances between schools and their nearest subway station. Future research should separate boroughs, as patterns may vary based on location.

Limitations

For this paper, I did not break results down by borough, though findings here indicated there may well be differences in results between boroughs. Future research should examine patterns by borough, rather than grouping NYC as a whole. I also lacked data on the level of transit use for each school's population; having that data would allow me to control for the percent of students who actually use the subways every day. It is possible that there are differences in the impact of subway variety and distance for commuter-heavy schools compared to those with a predominantly local population. My methods also do not account for schools that

share the same building that may differ in performance. Finally, I did not differentiate between tracts that are mostly inhabited land and those that are mainly composed of water for the purpose of mapping, though it is mainly cosmetic in this instance.

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